

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 2nd Term (Regular) Examination, 2020
Department of Materials Science and Engineering

MSE 3201
(Materials Characterization)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Calculate the structure factor for Ni_3Al unit cell and show five planes from where reflection may occur. (15)
- (b) Full width at half maximum, B of an X-ray diffraction peak of copper sample at $2\theta = 68^\circ$ is found 0.18° . Calculate the crystal size. X-ray diffraction test was conducted using $\text{Mo K}\alpha$ ($\lambda=0.721\text{\AA}$) radiation. (07)
- (c) "X-ray diffraction is an effective tool to examine residual stress in solids"-do you agree or disagree with the statement? Justify your answer. (08)
- Q 2. (a) Differentiate between backscattered electrons and secondary electrons in terms of their formation and role in image generation during SEM. (08)
- (b) "Long working distance (D_w) and small aperture radius (R_{fa}) is favorable for high depth of field but these will result in negative effects on the image resolution" why? What is your selection considering such tradeoffs between resolution and depth of field? (15)
- (c) "All materials are not susceptible to surface charging in SEM"-Explain. What is the effect of surface charging? (07)
- Q 3. (a) Explain how Bright Field and Dark Field images are formed in TEM. How a Scanning Tunneling Microscopy image is formed? (15)
- (b) You have been told that wedge, fringe and bending contours can be distinguished. Explain how. (15)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) How will you determine the composition of a thin film? Answer using technical details only. (08)
- (b) What are the differences between “standalone XRF” and “EDS in Electron Microscopes” (12)
- (c) “The machine atmosphere plays an important role in qualitative analysis”-do you agree or disagree with the statement? Justify your answer. (10)
- Q 5. (a) How will you analysis a polymeric sample using DTA? How does “SIMS” work (10+06)
- (b) Abbreviate “OES” and write down the basic principles of elemental analysis using OES. (14)
- Q 6. (a) What can you determine using vibrational spectroscopy techniques? How do these techniques work? (10)
- (b) Which spectroscopic technique is able to identify molecular bonds? How? (10)
- (c) How can you determine the composition of insulating materials using XPS? (10)

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MSE 3203
(Ceramics and Glass Engineering)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) State Pauling's rule 5 and point out why it is not applicable for all ceramic compounds. (05)
(b) Write down the defect equations for any Two of the following: (15)
(i) Incorporation of CaO in ZrO_2 .
(ii) Formation of Schottky defects in Cu_2O .
(iii) Dissolution of Sc_2O_3 in $CaTiO_3$ during synthesis, assuming Sc enters on Ti sites.
(iv) Frenkel defect in an ABO_3 compound.
(c) Zinc Sulphide (ZnS) has Zinc Blende structure. If the density is 3.02 g/cm^3 and the lattice parameter is 0.59583 nm , determine the number of Schottky defects per cubic cm. (10)
[Atomic wt. of Zinc = 65.38 g/cm^3 , Atomic wt. of Sulphur = 32 g/cm^3]
- Q 2. (a) How does temperature and strain rates effect the creep curve of ceramics? (10)
(b) A series of test on a ceramic component indicate that 25% of the samples fail at stresses below 207 MPa , whereas, only 10% fail at stresses below 198 MPa . Determine the Weibull Modulus and the maximum allowable stresses if only 1 failure per 1000 specimens can be tolerated. (10)
(c) 'The size and shape of crack significantly effects the fracture behavior of a ceramic'- (10)
Explain based on the following equation:

$$\frac{\sigma_m}{\sigma_s} = 2 \left(\frac{C}{\rho} \right)^{\frac{1}{2}}$$

Where, σ_m = maximum stress at orack tip. σ_s = applied stress

$2C$ = crack length, ρ = radius of crack tip.

- Q 3. (a) Ceramics show lower strength than their estimated theoretical strength, why? (05)
(b) What is modulus transfer and how does it toughen ceramics? (10)

(c) For the Table 3(c)

(15)

Ceramic	E, GPa	K_{IC} , MPa.m ^{1/2}
(A) ZrO ₂ (cubic)	220	3.0-3.6
(B) ZrO ₂ (partially stabilized)	190	3.0-15.0

Why does (B) show a greater range of fracture toughness data? Explain.

Section B

(Answer ANY TWO questions from this section in answer script B)

- Q 4. (a) Define energy band and how does it form for Si ? (15)
- (b) Explain the effect of temperature on insulator, semiconductor, conductor and superconductor. (15)
- Q 5. (a) Why powder reactivity is poor in solid state reactions method? (07)
- (b) What are the criteria should maintain for better reactions of reactants in solid state reaction method. (11)
- (c) What are the problems that can occur during uniaxial Die pressing and how these problems can overcome? (12)
- Q 6. (a) Describe wet bag and dry bag isostatic pressing process. (15)
- (b) Write a short note on YBCO superconductor. (07)
- (c) Why calcination step absent in hydrothermal method for the synthesis of nanoparticles? (08)

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MSE 3205
(Corrosion and Surface Engineering)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Explain why 18-8 stainless steel susceptible to intergranular corrosion at elevated temperature. Briefly discuss the method for minimizing intergranular corrosion of 18-8 steel. (18)
- (b) What is Galvanized steel? Which type of protection is used here, Anodic or Cathodic? Please explain it with respect to the scenario of Bangladesh. (12)
- Q 2. (a) Discuss the autocatalytic nature of Pitting corrosion using schematic diagram. (10)
- (b) Explain the relationship between Pit Depth and Number of Pits appearing on the corroded surface. (10)
- (c) "Metal wants to corrode" – Explain. (10)
- Q 3. (a) Enamelled surfaces of food and beverage cans often exposed to atmosphere and Red Brown corrosion filaments are visible. The corrosion filaments consist of an active head and an inactive tails. Explain the corrosion initiation mechanism with propagation. (10)
- (b) Relate Pitting and Crevice corrosion. (10)
- (c) What type of corrosion would take place on a sheet of iron that is placed on top of a roof? Write down the preventive measures that should be taken to protect this iron sheet. (10)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) Explain Sputtering process. (12)
- (b) How Sputtering process overcome the problems that related to evaporation process? (08)
- (c) Explain the factors that affect electroplating process. (10)
- Q 5. (a) Explain the steps of CVD process. Why CVD process is more convenient than PVD process? (9+7)
- (b) Why AC current is not used in electroplating process? (07)
- (c) Explain why sodium silver cyanide electrolyte is preferred to silver nitrate for Ag electroplating? (07)
- Q 6. (a) Explain delamination theory of wear. (10)
- (b) What are the problems that caused by adhesive wear? How to avoid abrasive wear? (12)
- (c) Explain three-body abrasive wear mechanism. (08)

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MSE 3209
(Extractive Metallurgy)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) A sample of bauxite consists of 10, 25, 15, 30, 20, 5 μm particles. Analyze this sample using the Warman cyclosizer method with a suitable diagram. (15)
- (b) Draw a schematic diagram and explain the working principle of the Warman cyclosizer. (15)
- Q 2. (a) What is meant by comminution? Briefly explain the different grinding mechanisms with necessary figures. (15)
- (b) Consider a mixture of galena (specific gravity 8.3) and quartz (specific gravity 3.1) particles classifying in water. Now prove that free settling ratio is larger for coarse particles obeying Newton's law than fine particles obeying stoke's law. (15)
- Q 3. (a) What is froth flotation in ore dressing? (05)
- (b) Explain the basic principle of the centrifugal sedimentation dewatering technique with a suitable diagram. (15)
- (c) Why is the disposal of tailings so important in mineral processing? (10)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) What are the different types of roasting techniques in pyrometallurgy route? Please narrate the similarities and dissimilarities between oxidation Roasting and Volatilizing Roasting. (12)
- (b) In the practice of industrial roasting method one type of roasting route is fluidized bed roasting. Why so it is called as Fluidized bed? Please explain it with its mechanism. (12)
- (c) What are advantages and disadvantages of Fused Salt Electrolysis? (06)
- Q 5. (a) What is electrometallurgy? What are the important considerations that one should consider during extraction of pure metal? (10)
- (b) Please state Faradays Law of Electrolysis. Explain the mechanism that how electro-refining can be done by electrolysis process with suitable example and diagram. (15)
- (c) What is called current efficiency? (05)
- Q 6. (a) What are the steps involved in Hydro-metallurgy processing? Narrate few advantages of Hydrometallurgy over Pyrometallurgy route. (15)
- (b) What are techniques that involved in In Situ Leaching? Please dictate few difficulties in that involved during In Situ Leaching. (15)

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IEM 3227
(Industrial Management)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Discuss about the relationship between management and administration. (10)
(b) Briefly describe some factors of dissatisfaction in workplace. (08)
(c) Briefly discuss about the break-even point with suitable example. (12)
- Q 2. (a) Define management. Briefly explain about the fundamental functions of management. (10)
(b) What is path-goal theory? Explain with a suitable example. (10)
(c) Bad communication may harm the relation between leader and employee. How? (10)
- Q 3. (a) Why a good wage plan needs to be linked to employee performance? Explain. (10)
(b) What are the factors that determine wage and salary structures? Explain briefly. (10)
(c) Where the piece wage system should be used? (10)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) How does 'Native Forecasting' work? (05)
(b) Healthy hamburger has a chain of 12 stores in northern Illinois. Sales figures and profits for the stores are given in the following table. Obtain a regression line for the data and predict profit for a store assuming sales of \$10 million. (15)

Unit sales, x	7	2	6	4	14	15	16	12	14	20	15	7
Profits, y	0.15	0.10	0.13	0.15	0.25	0.27	0.24	0.20	0.27	0.44	0.34	0.17

Both sales and profits are given in \$ millions.

- (c) What are the basics strategies to meet uneven demand? Briefly explain them. (10)
- Q 5. (a) The product structure tree for end item E is given in the following figure 5(a). The manager wants to know the material requirements for ordered part R that will be needed to complete 120 units of E by the start of week 5. Lead times for items are one week for level 0 items, one week for level 1 items and two weeks for level 2 items. There is scheduled receipt of 60 units of M at the start of week 2 and 100 units of R at the start of week 1. Lot-for-lot ordering is used. (18)

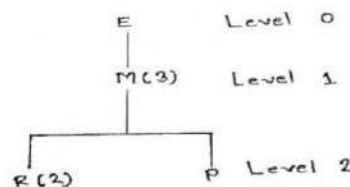


Figure: 5 (a)

- (b) Surge electric uses 4,000 toggle switches a year. Switches are priced as follows: 1 to 499, 90 cents each; 500 to 999, 85 cents each and 1,000 or more, 80 cents each. It costs approximately \$30 to prepare an order and receive it, and carrying costs are 40 percent of purchase price per unit on annual basis. Determine the optimal order quantity and total annual cost. (12)

- Q 6. (a) A group of six jobs is to be processed through a two machine flow shop. The first operation involves cleaning and the second involves painting. Determine a sequence that will minimize the total completion time for this group of jobs. Processing times are as follows: (15)

Job	Work Centre 1	Work Centre 2
A	5	5
B	4	3
C	8	9
D	2	7
E	6	8
F	12	15

Show the sequence by constructing Gantt chart to reveal flow time and idle time.

- (b) Expected time and variance for the major activities of an R & D project are depicted in the following PERT chart (Figure 6 (b)). Determine the probability that project completion time will be (15)

i) 50 weeks or less

ii) More than 50 weeks

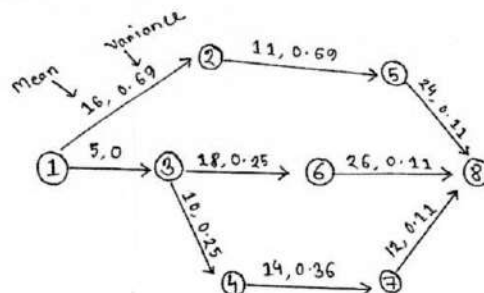


Figure: 6 (b)